

Analog to Digital Converters (ADC)

ANALOG DIGITAL CONVERTERS

Analog-to-Digital Converter (ADC)

- A device that converts continuous analog signals to a discrete digital signals.
- Conversion process is done via quantization (mapping a large set of inputs into a smaller set) and therefore a small amount of error is introduced. Conversion is periodical, ADC samples the input signal.
- This feature can be achieved through a few different ways but the most common technique is charging a capacitor and measuring the time it takes to discharge across an internal resistor. The controller counts the clock cycles and returns the count as the digital voltage level.
- This level is represented as binary on a number of bits according to the ADC resolution. For example, a 10-bit ADC has the ability to detect 1024 (2^{10}) discrete analog levels.
- After the conversion, ADC reports a ratiometric value; e.g. if an ADC works on a 5V supply, the maximum discrete level (1023 in 10-bit resolution case) responds to 5V.

$$\frac{ADC \text{ Resolution}}{System \text{ Voltage}} = \frac{ADC \text{ Reading}}{Measured \text{ Analog Voltage}}$$

- Therefore, if an analog voltage of 2.12V is measured at a system working on a 5V voltage supply with an ADC of 10-bit resolution ($2^{10} = 1024$):

$$\frac{1023}{5V} = \frac{ADC \text{ Reading}}{2.12V} \Rightarrow ADC \text{ Reading} = 434 \text{ LSB (Least Significant Bit)}$$

- and this value is reported to the microcontroller registers as 10-bit binary: 0110110010

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Input Voltage in LSBs for a 10-bit ADC ($n = 10$):

VIN in LSBs = VIN in Volts / $((2^{10} - 1) \times 5V)$

$2^{10} = 1024$ number of Quantization Steps

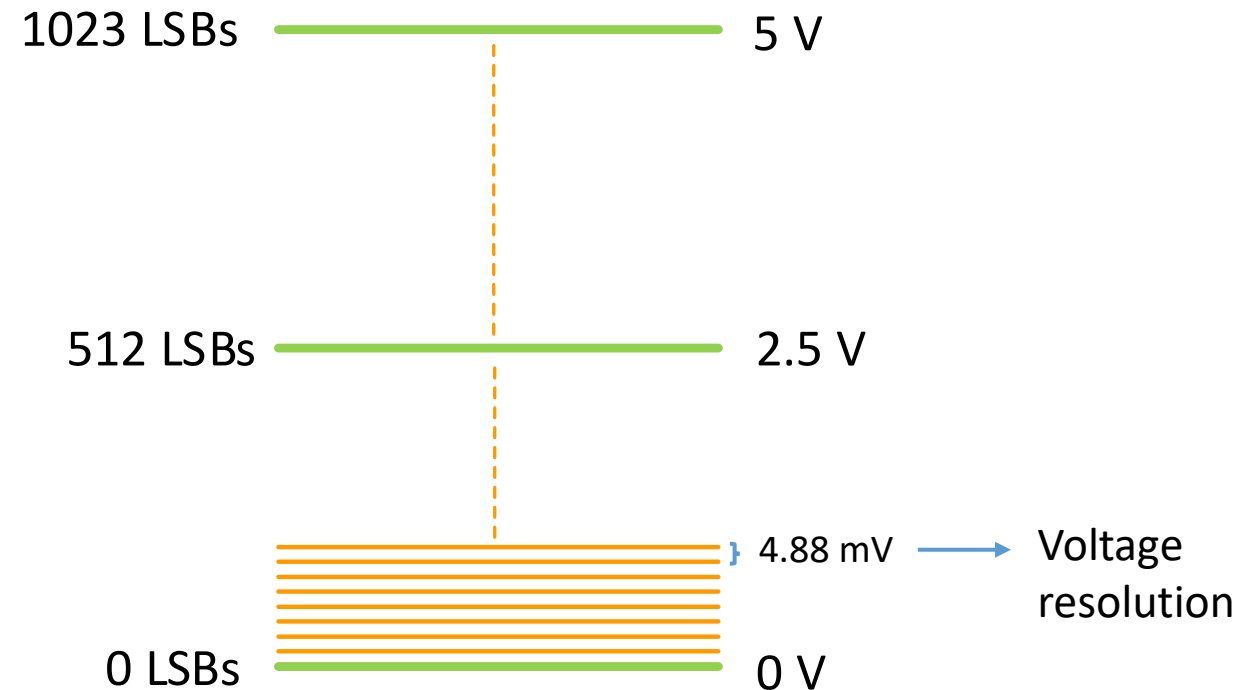
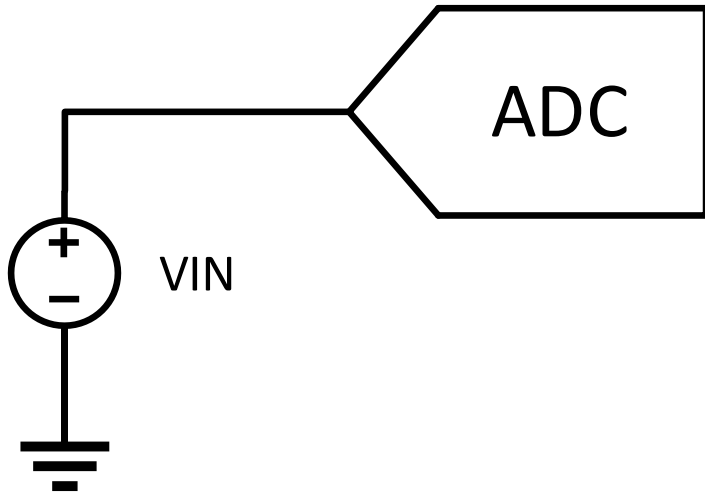
Resolution in Volts = $V_{IN} / (2^{10}-1)$

Smallest voltage that can be detected by ADC

Example: If $V_{IN} = 10 \text{ mV}$, what would be the conversion result?

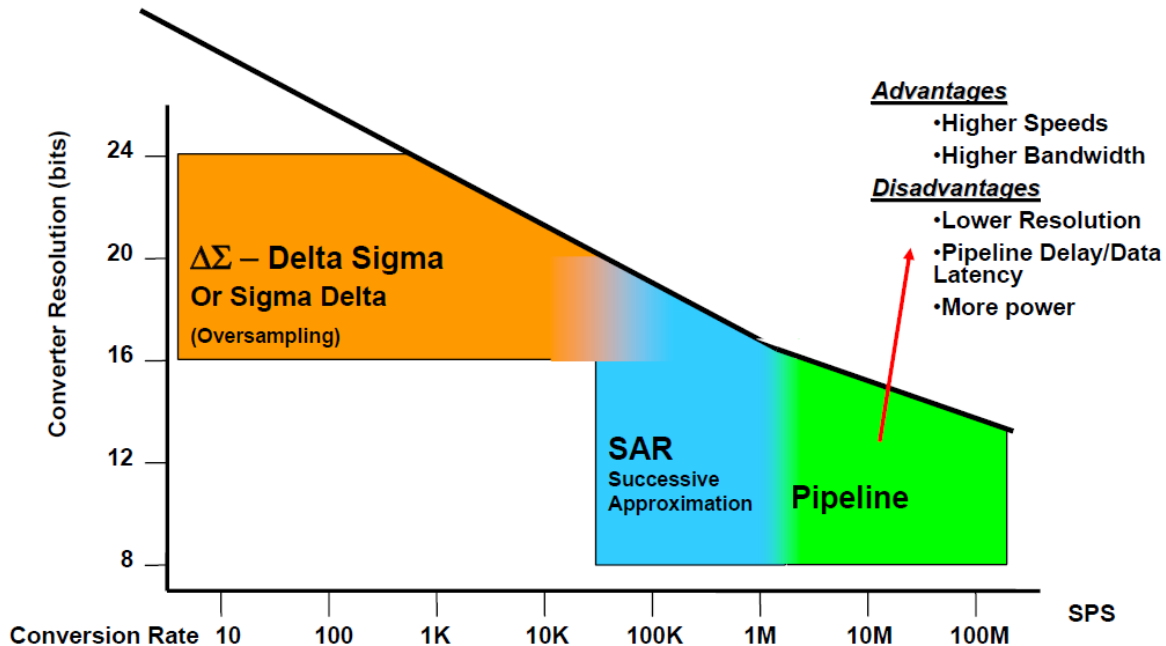
Answer: 2 LSBs $\times 4.88 \text{ mV} = 9.76 \text{ mV}$

Error = $10 \text{ mV} - 9.76 \text{ mV} = 0.24 \text{ mV} \rightarrow \text{Quantization Noise!}$



ANALOG DIGITAL CONVERTERS

ADC Types:



Depending on the application, a trade-off between speed and resolution can be made

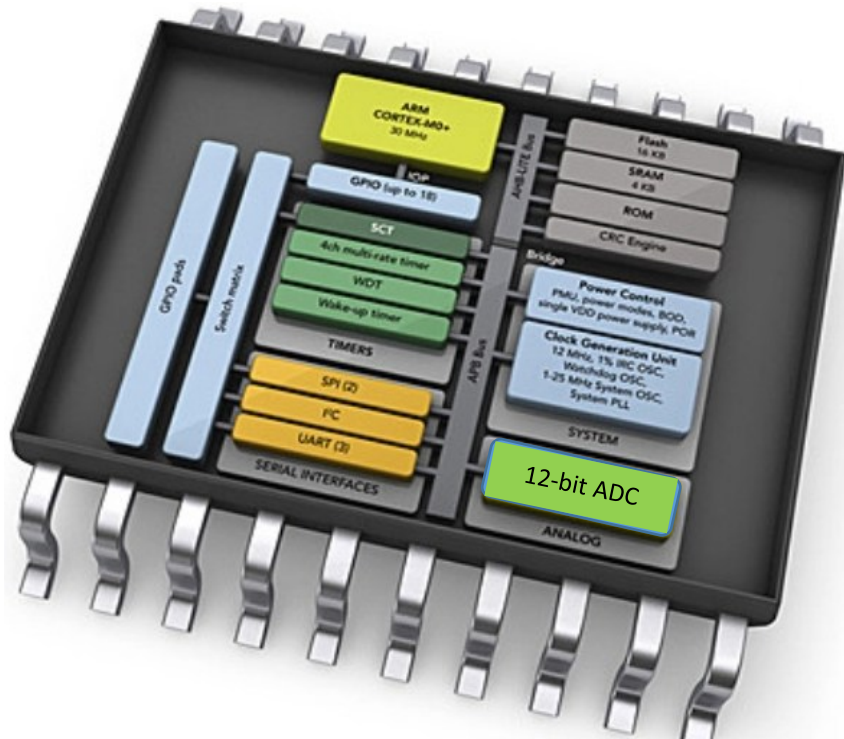
A **SAR (Successive Approximation Register) ADC** converts an analog signal into a digital value by comparing it step-by-step with a reference using a binary search algorithm.

It offers a good balance of speed, resolution, and low power, which makes it the most widely used ADC type in microcontrollers and embedded systems. SAR ADCs are commonly found in automotive ECUs, industrial controllers, and IoT devices due to their deterministic conversion time and simplicity.

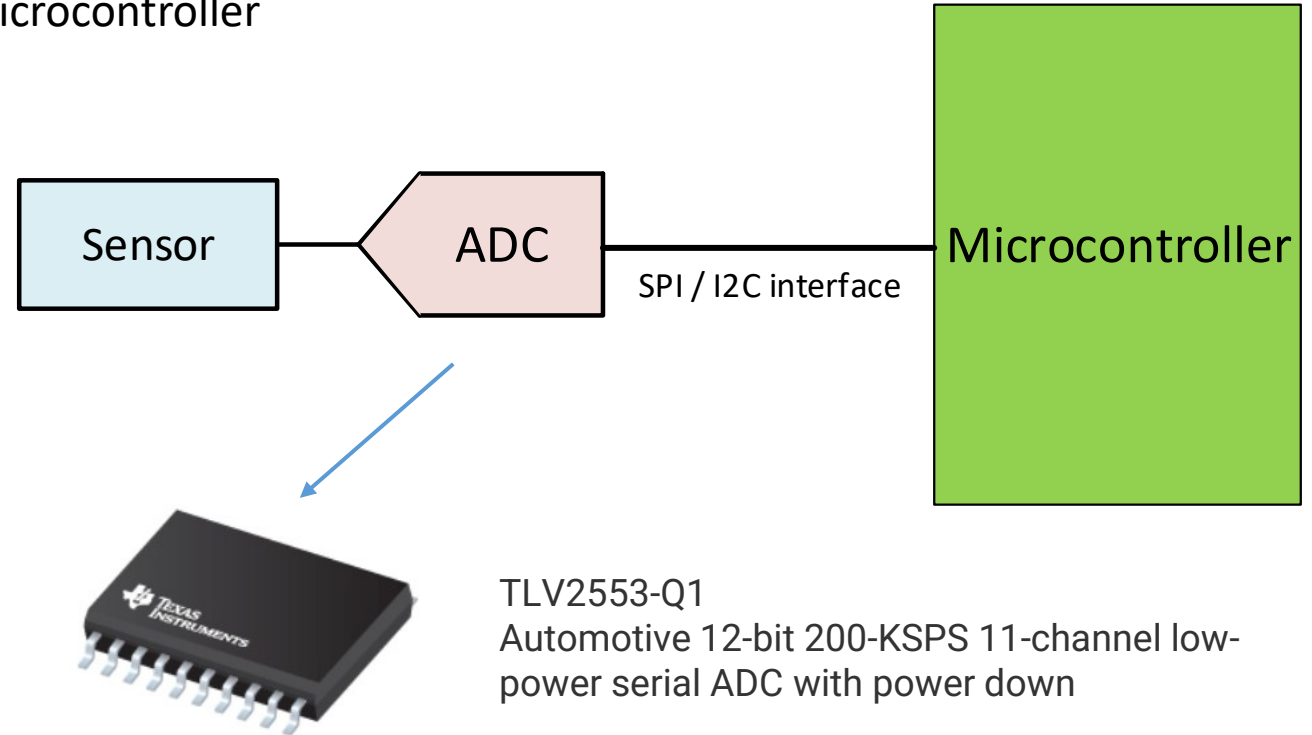
<https://www.analog.com/en/analog-dialogue/articles/the-right-adc-architecture.html>

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ADC can be either standalone or integrated in the microcontroller



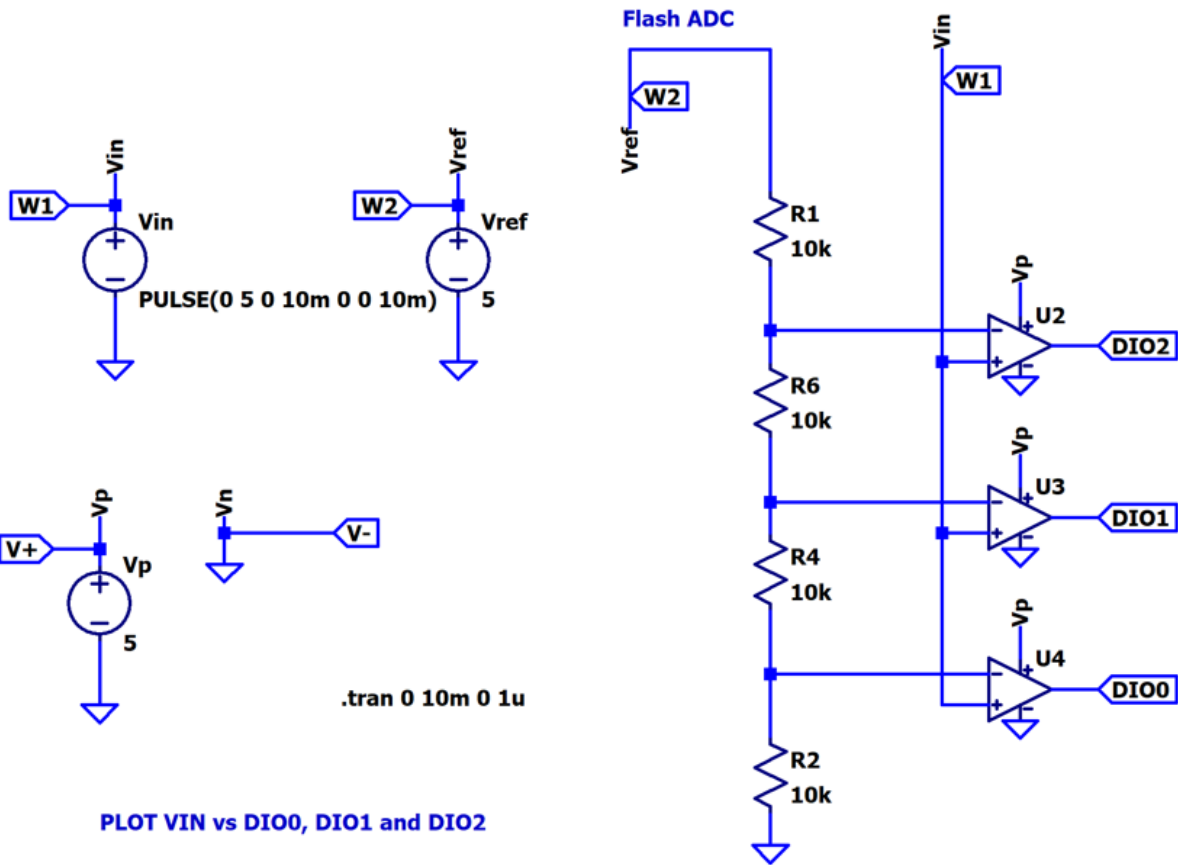
ADC integrated in the microcontroller – NXP LPC8XX
low cost microcontroller family



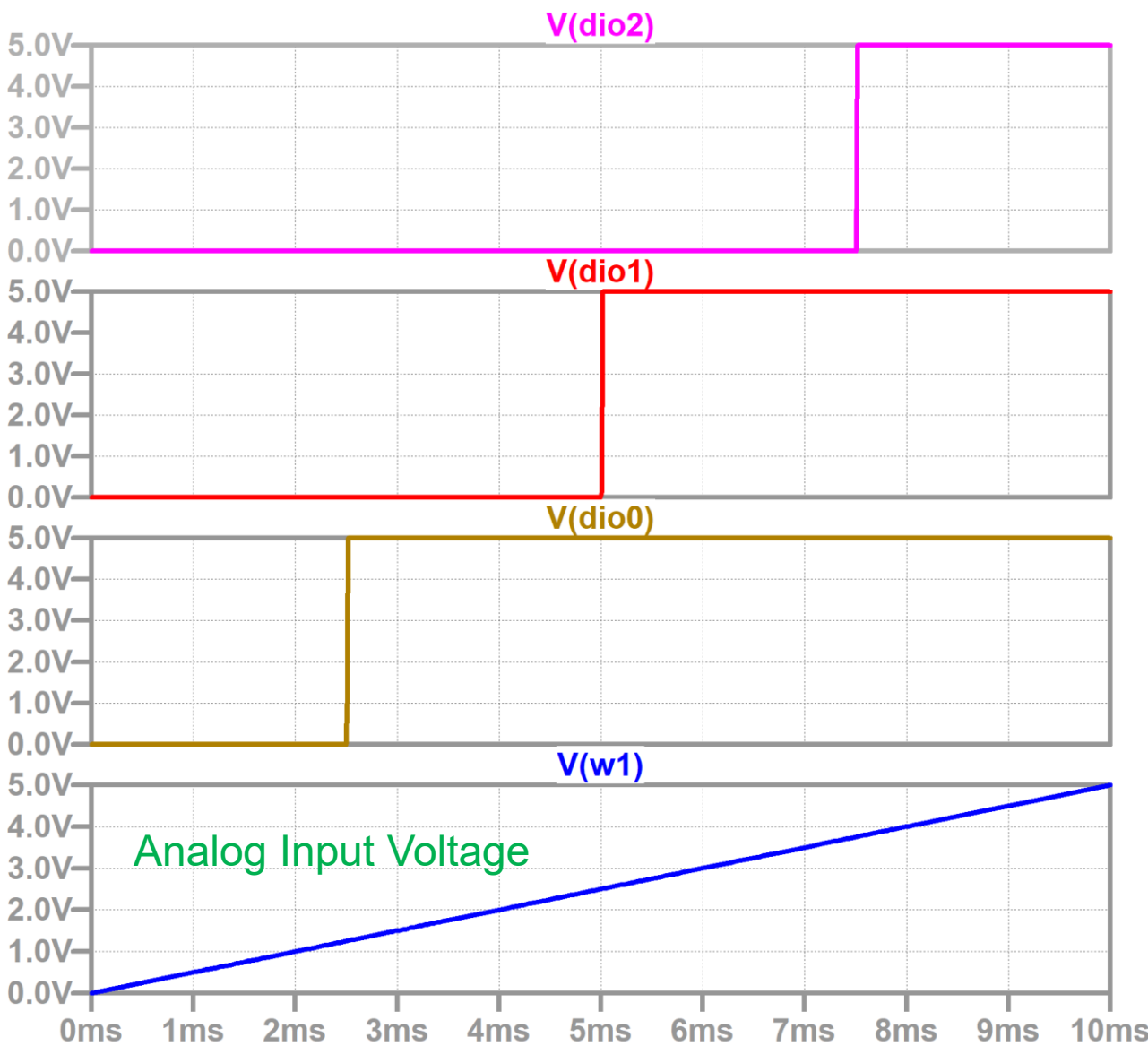
An external ADC can be preferred if

- the microcontroller does not have enough ADC inputs or
- the integrated ADC in the microcontroller does not fulfill the performance requirements (resolution, speed etc.)

FLASH CONVERTER ADC

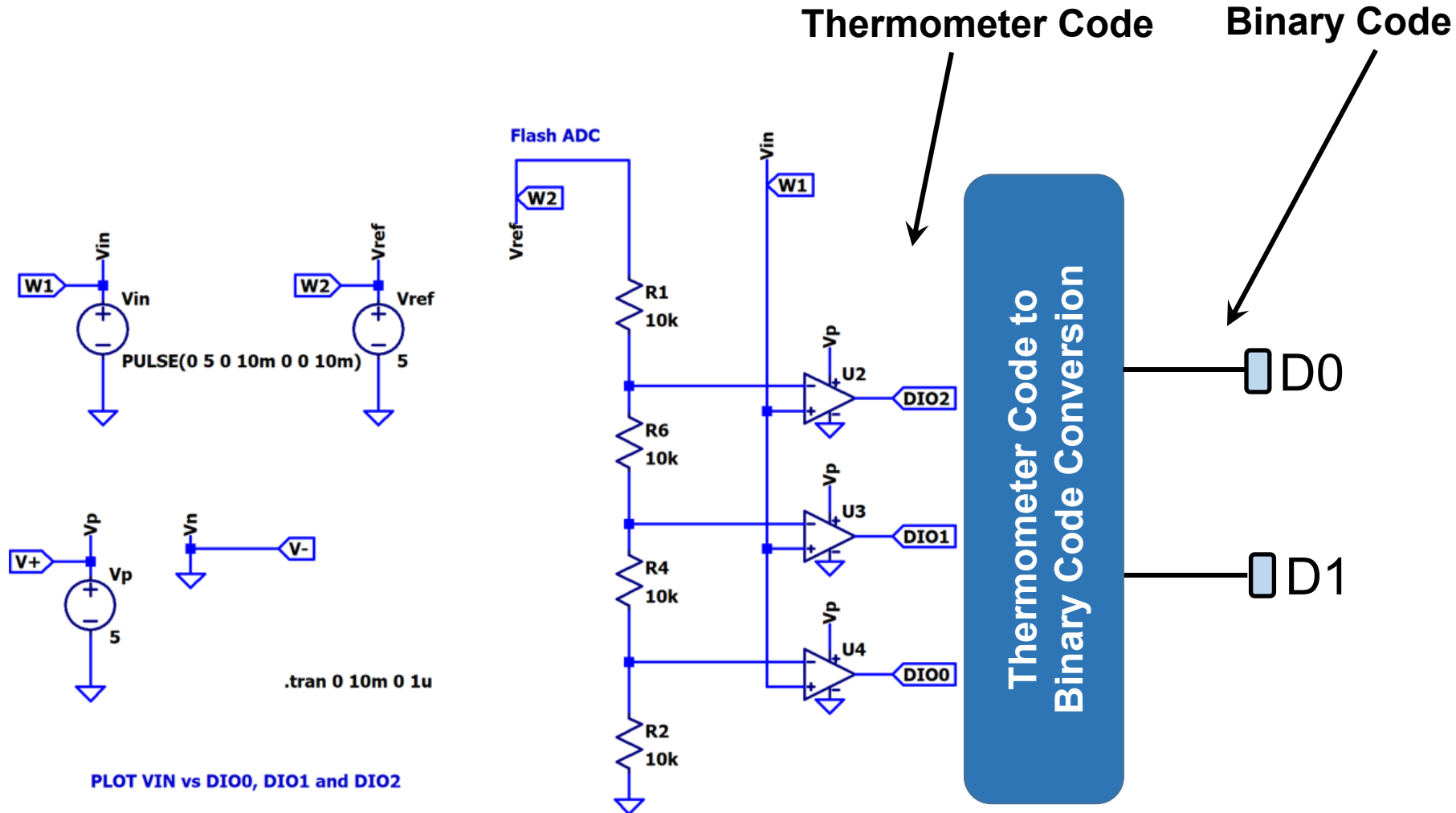


Basic Flash ADC Circuit



Analog Input Voltage vs Digital Thermometer Code
Digital Output

FLASH CONVERTER ADC



Thermometer				Binary	
DIO0	DIO1	DIO2	DIO3		
0	0	0	0	→	0 0
0	0	0	1	→	0 1
0	0	1	1	→	1 0
0	1	1	1	→	1 1

For 2-bit resolution, 3 comparators would be enough ($2^N - 1$), since DIO0 is always 0!

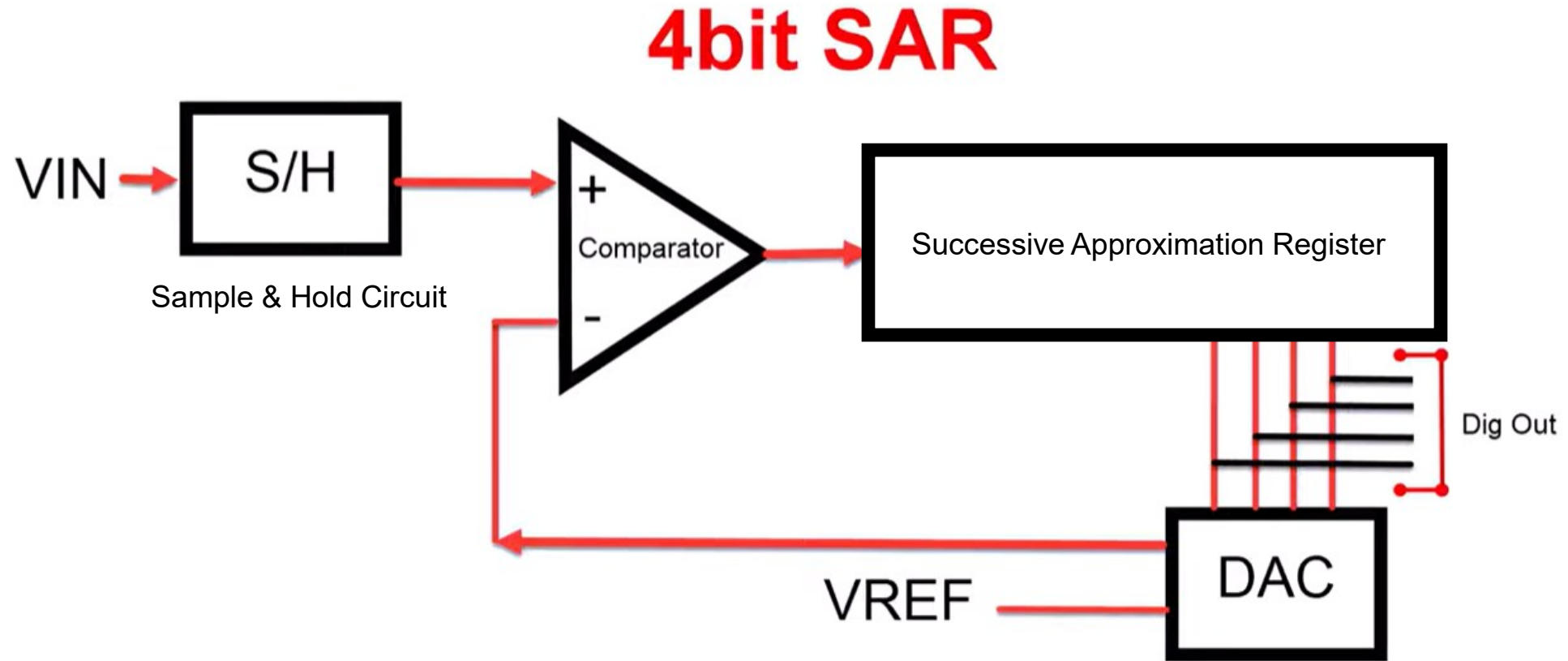
Basic 2-bit Flash ADC Circuit

FLASH CONVERTER AND SAR ADC

A flash converter is a very fast type of ADC; however, achieving high resolution requires a large number of comparators (for 2 bits → at least 3 comparators). This significantly increases chip area and power consumption.

Therefore, in these lecture notes, we will focus on the SAR ADC, since it is the most widely used ADC in practical applications and offers a good compromise between speed and resolution.

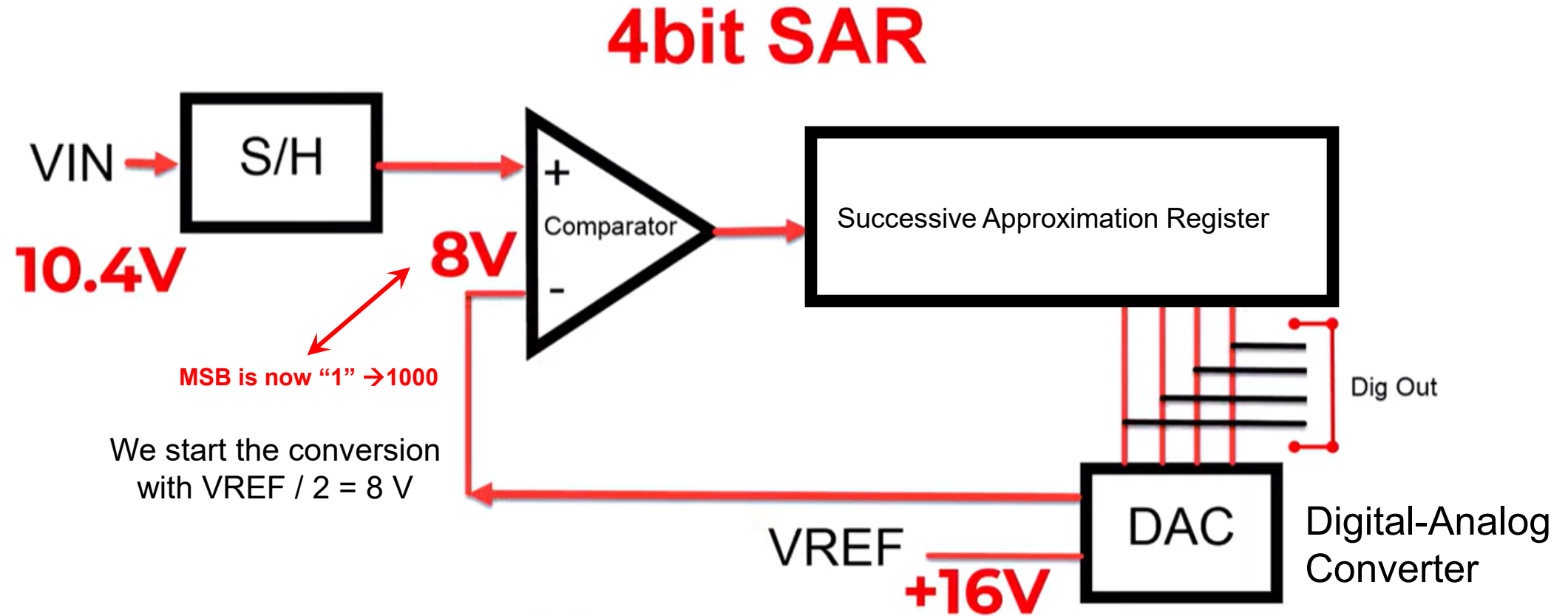
SAR ADC



Basic 4-bit SAR ADC topology → we now have only one comparator!

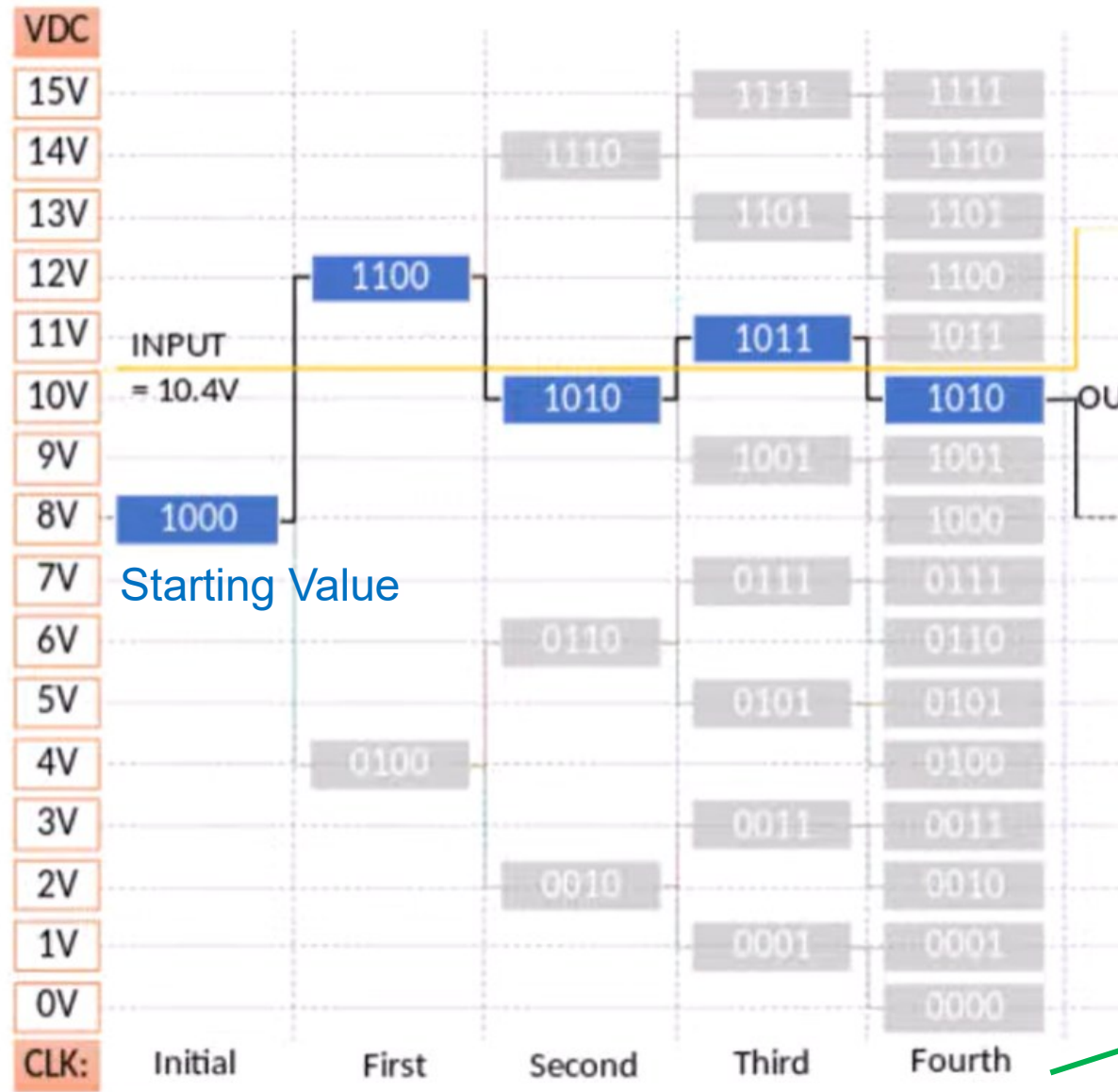
SAR ADC

$$V_{\text{REF}} = 16 \text{ V} \rightarrow \text{constant}$$



These voltages are not realistic because the maximum input voltage of most of the SAR ADCs is 5.0 V. For the sake of simplicity however, we chose these voltages to explain the working principles of SAR ADCs

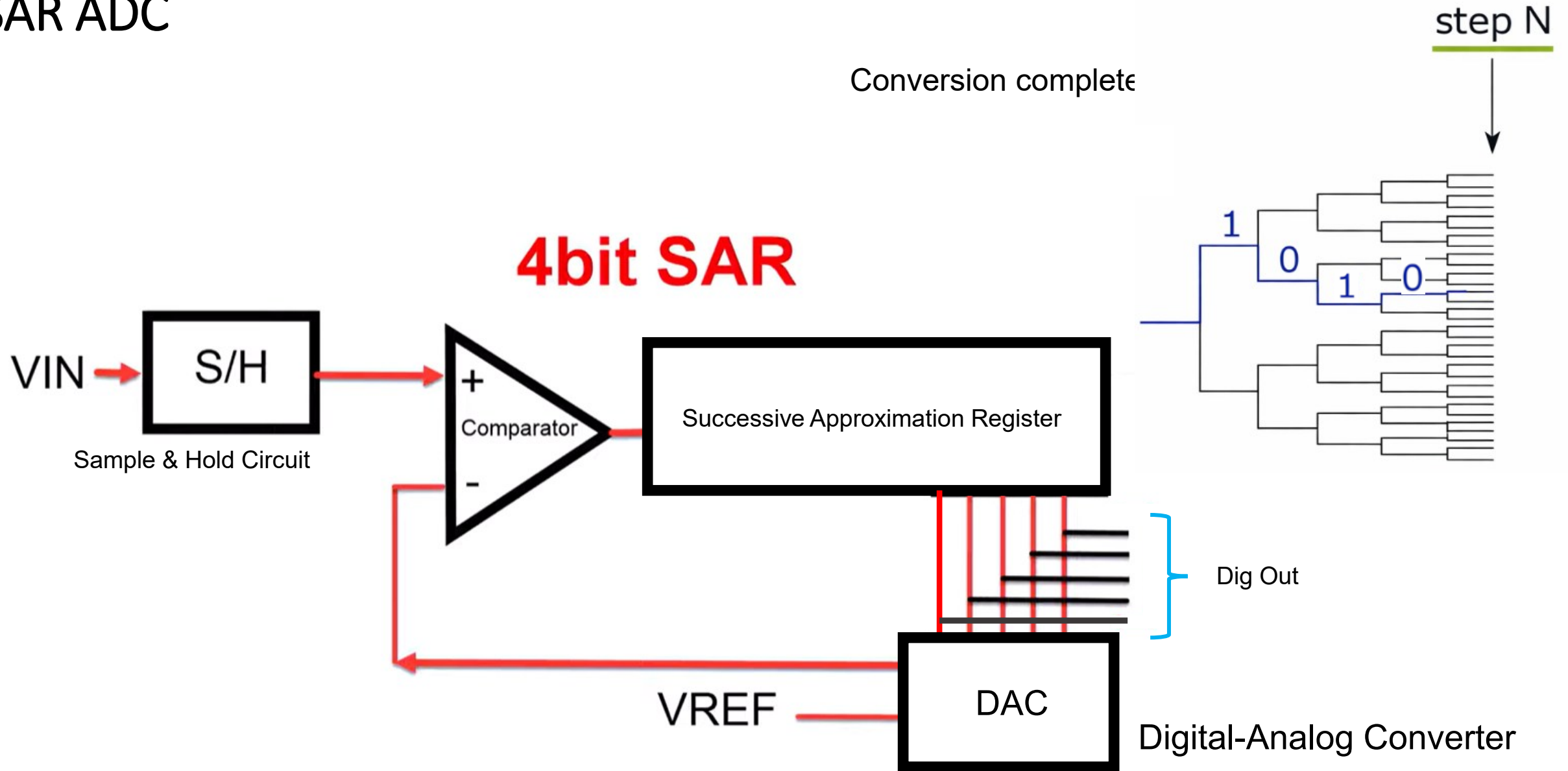
SAR ADC



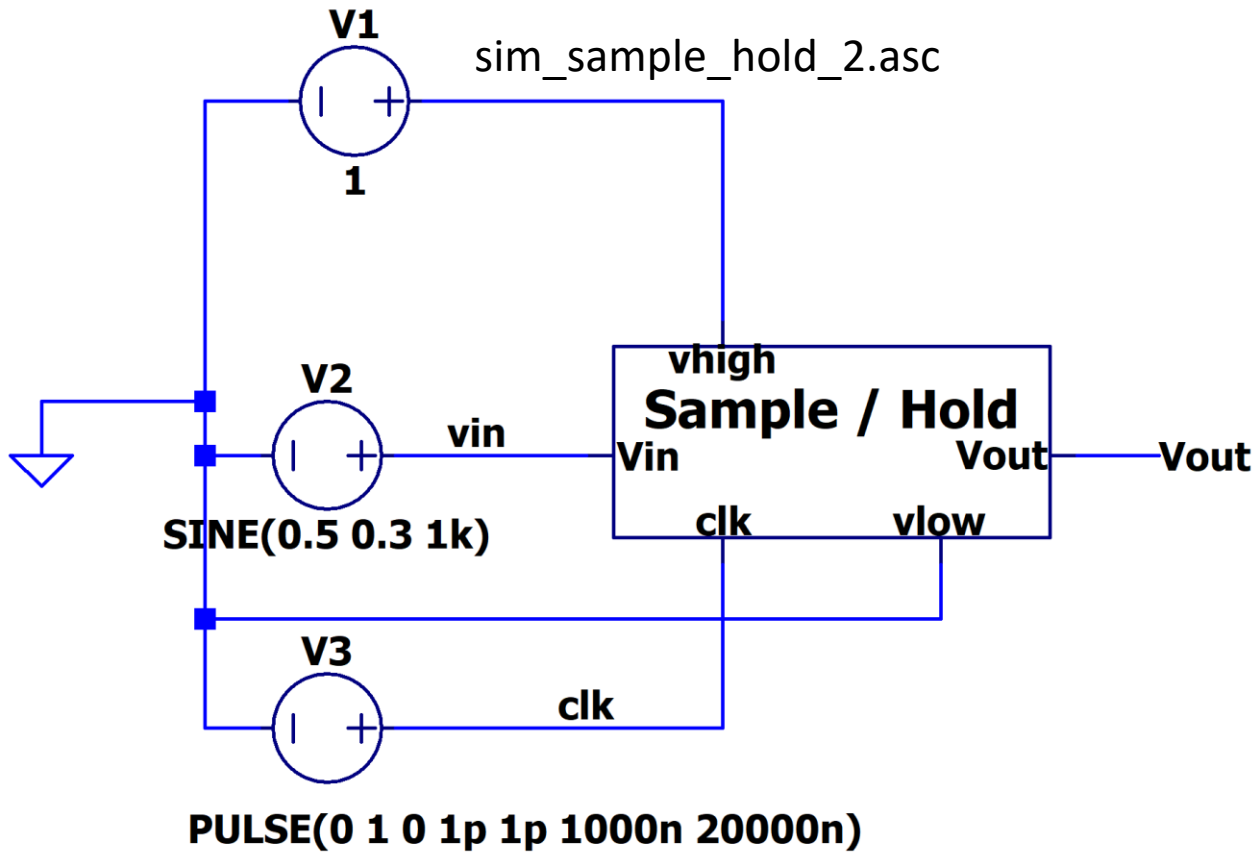
Conversion Result = 10 V

4-bit SAR ADC finishes the A-D conversion in four clock cycles!

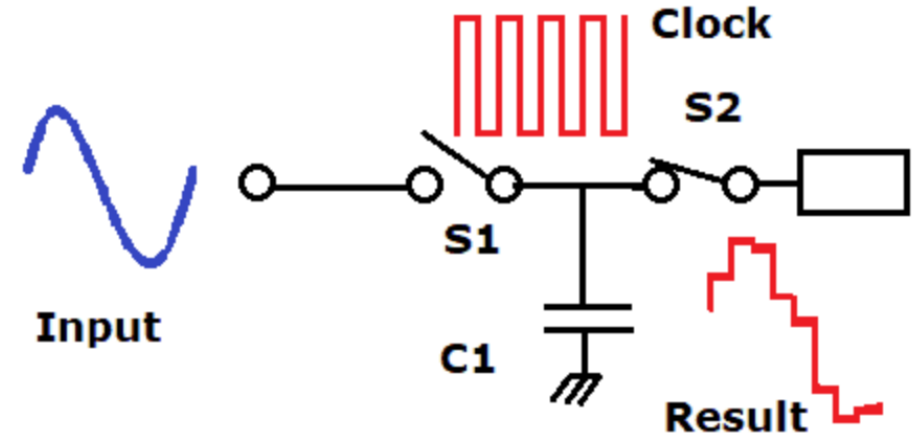
SAR ADC



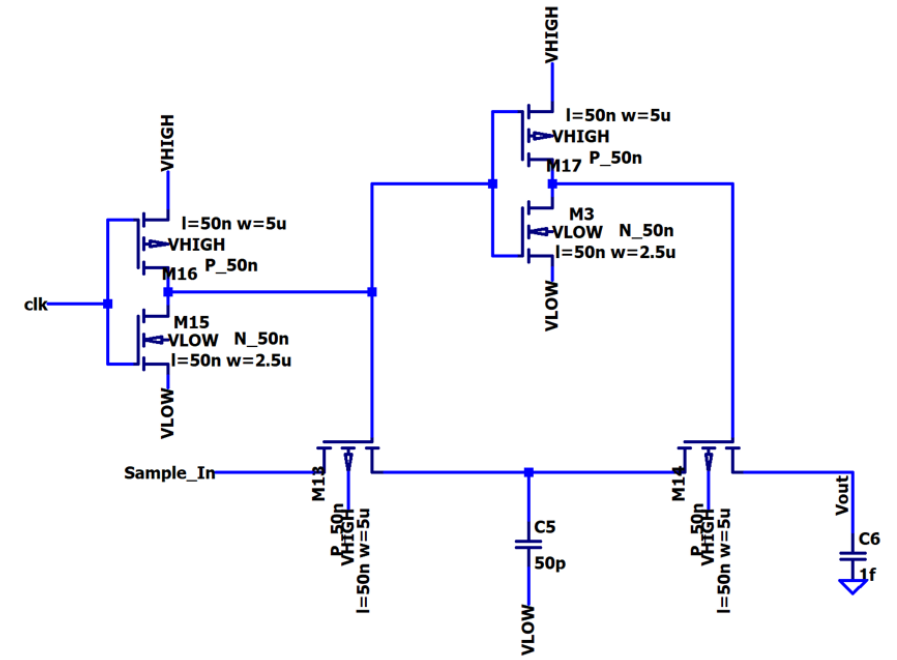
SAR ADC – SAMPLE AND HOLD CIRCUIT



```
.include models.txt
.tran 1m
.options plotwinsize=0
```

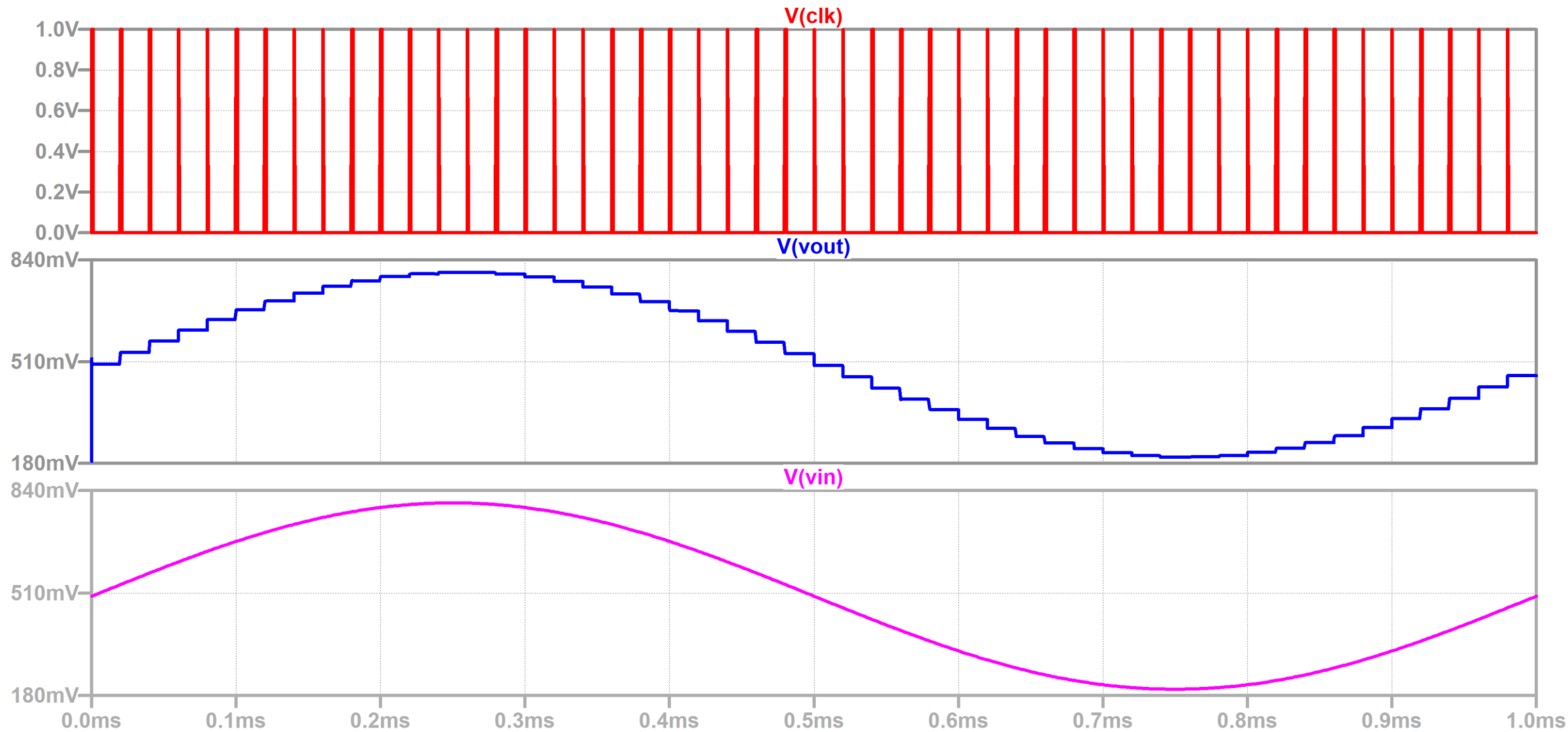


Simplified Equivalent Circuit of Sample & Hold Circuit

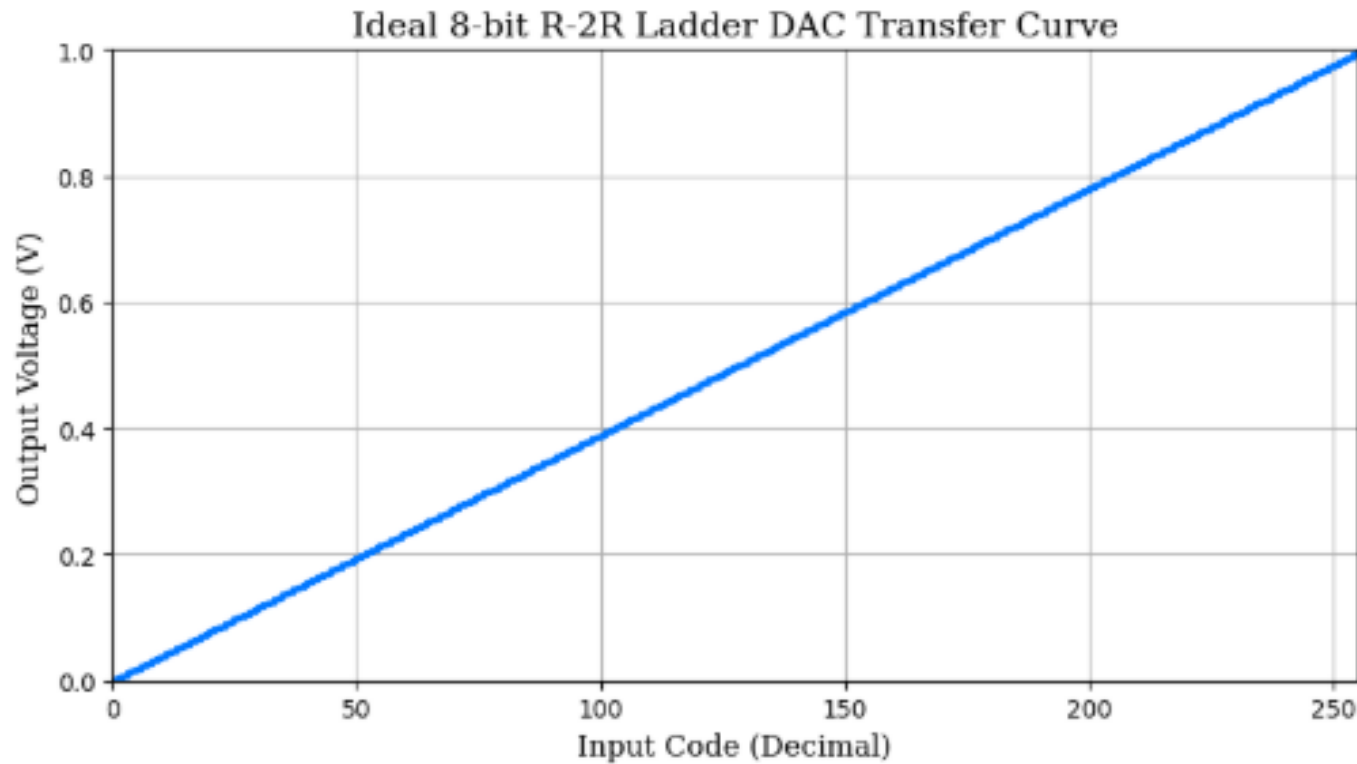
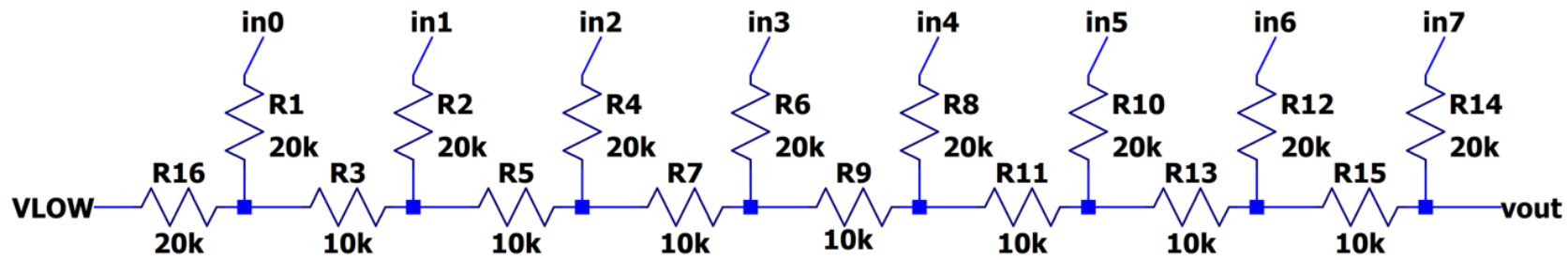


Sample & Hold Circuit → *sample_hold_2.asc*

SAR ADC – SAMPLE AND HOLD CIRCUIT

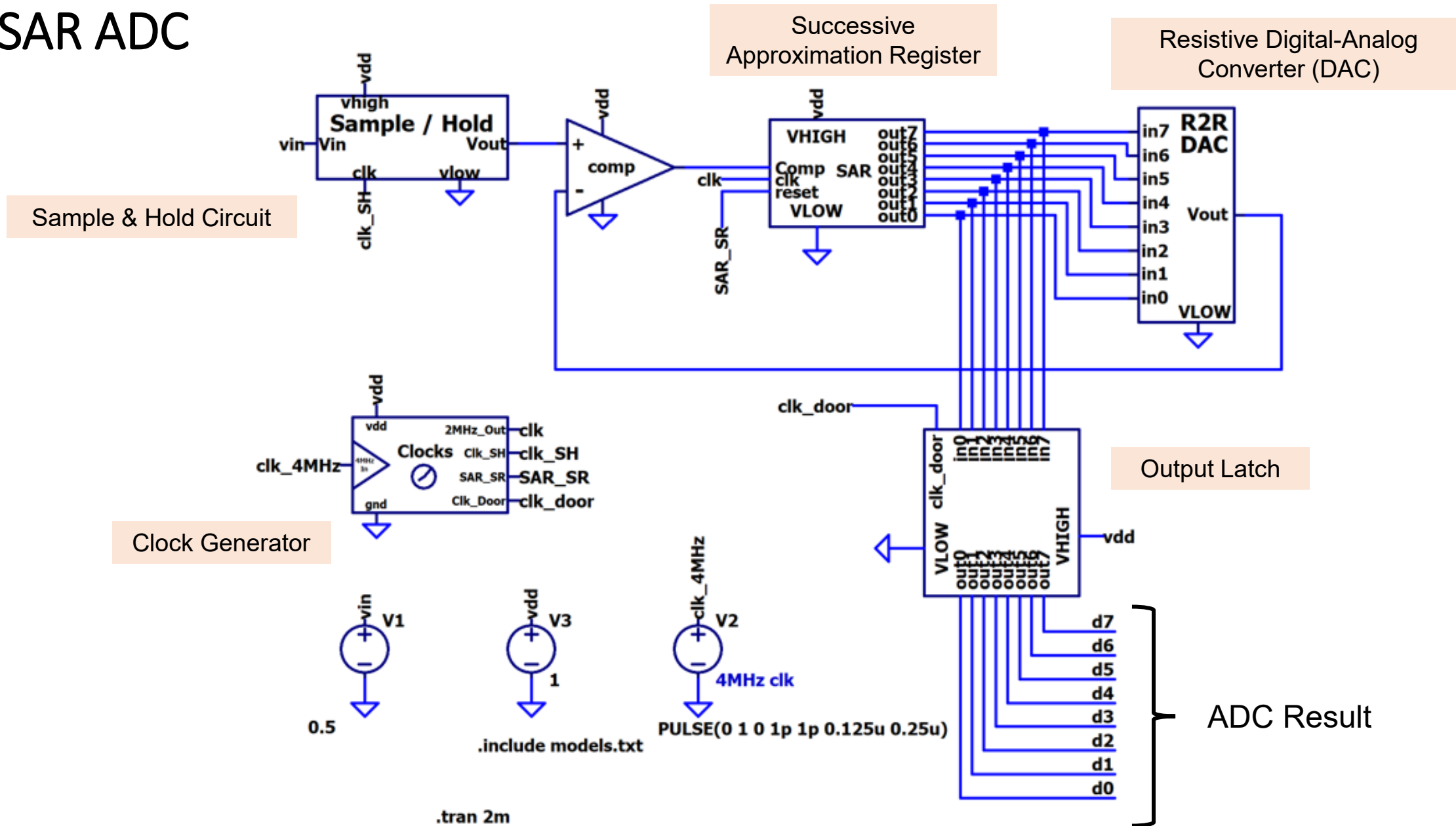


SAR ADC – R2R LADDER DAC



8-bit → 256

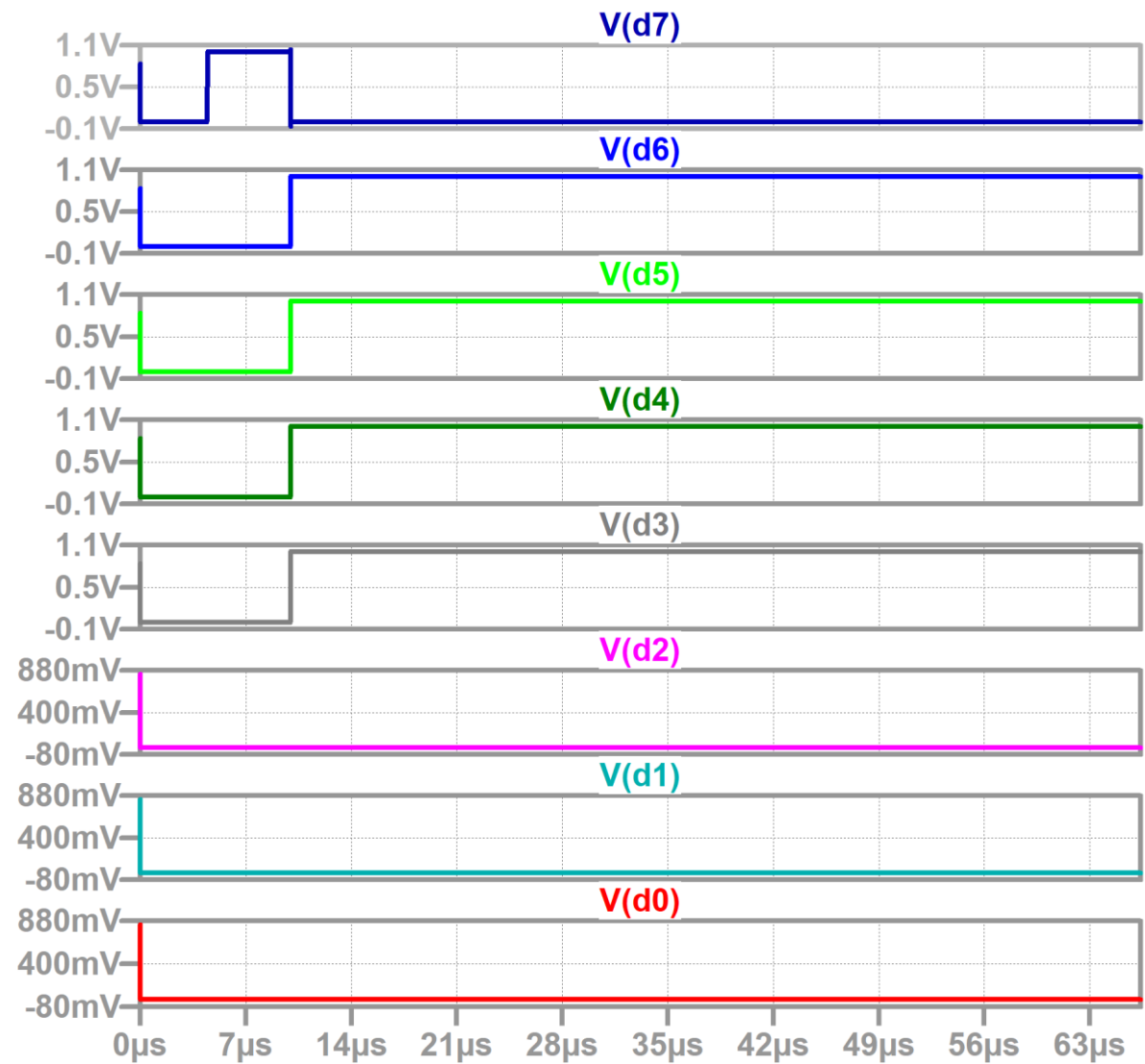
SAR ADC



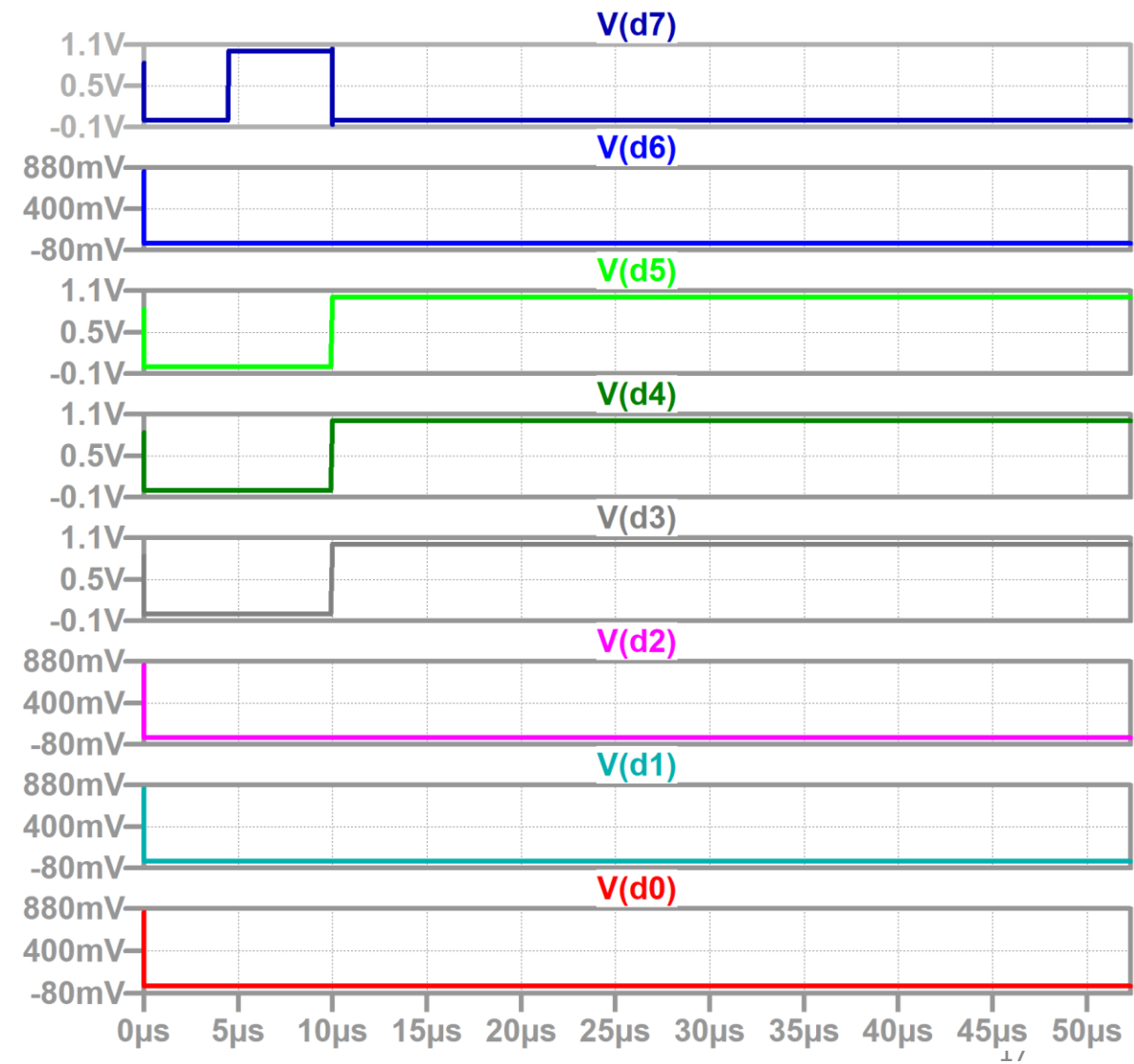
sim_SAR_ADC_simple.asc

SAR ADC – EXAMPLES

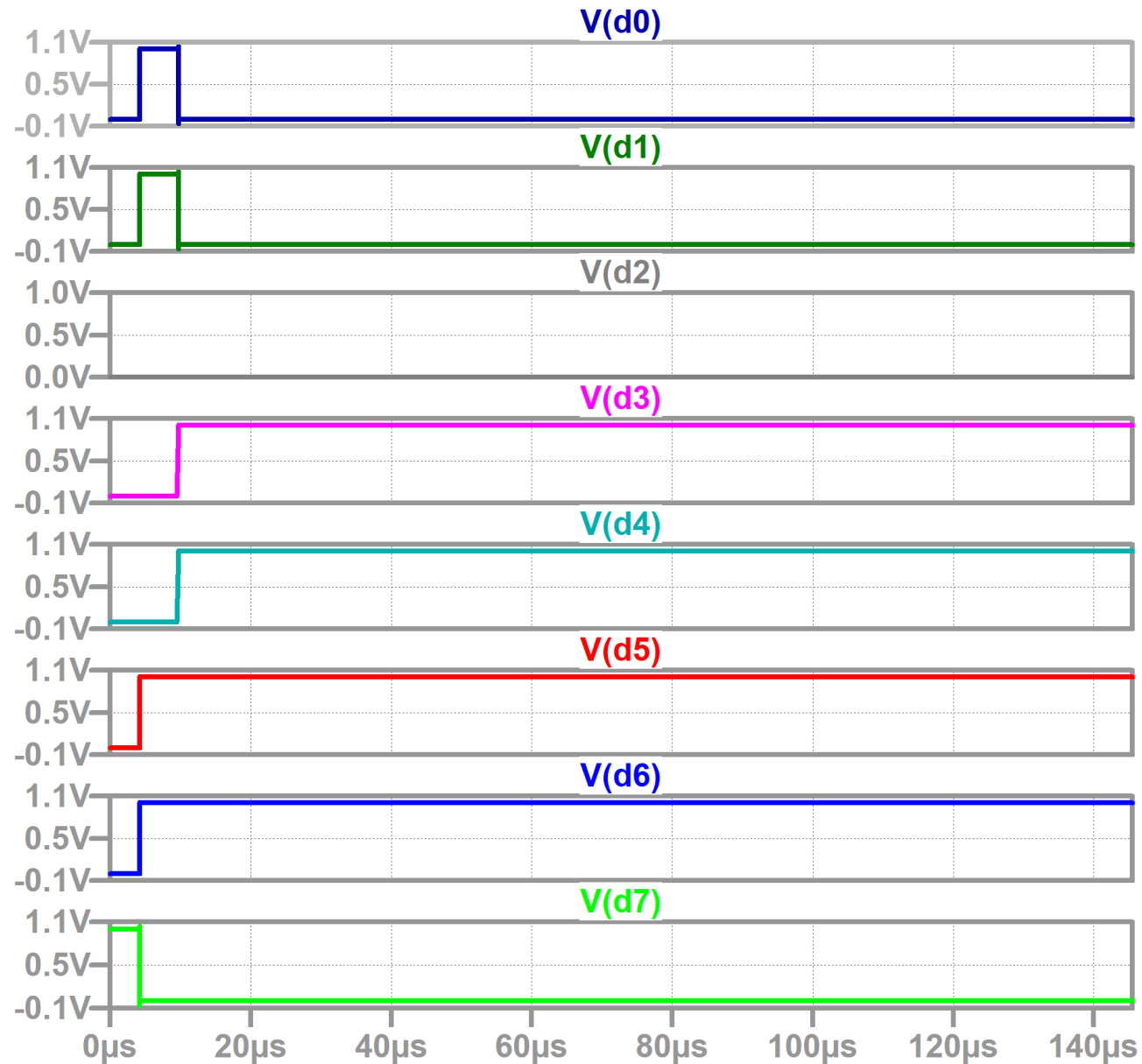
$V_{in} = 0.5\text{ V}$, $V_{ref} = 1\text{ V}$



$V_{in} = 0.25\text{ V}$, $V_{ref} = 1\text{ V}$



SAR ADC – INPUT DC = 0.5 V



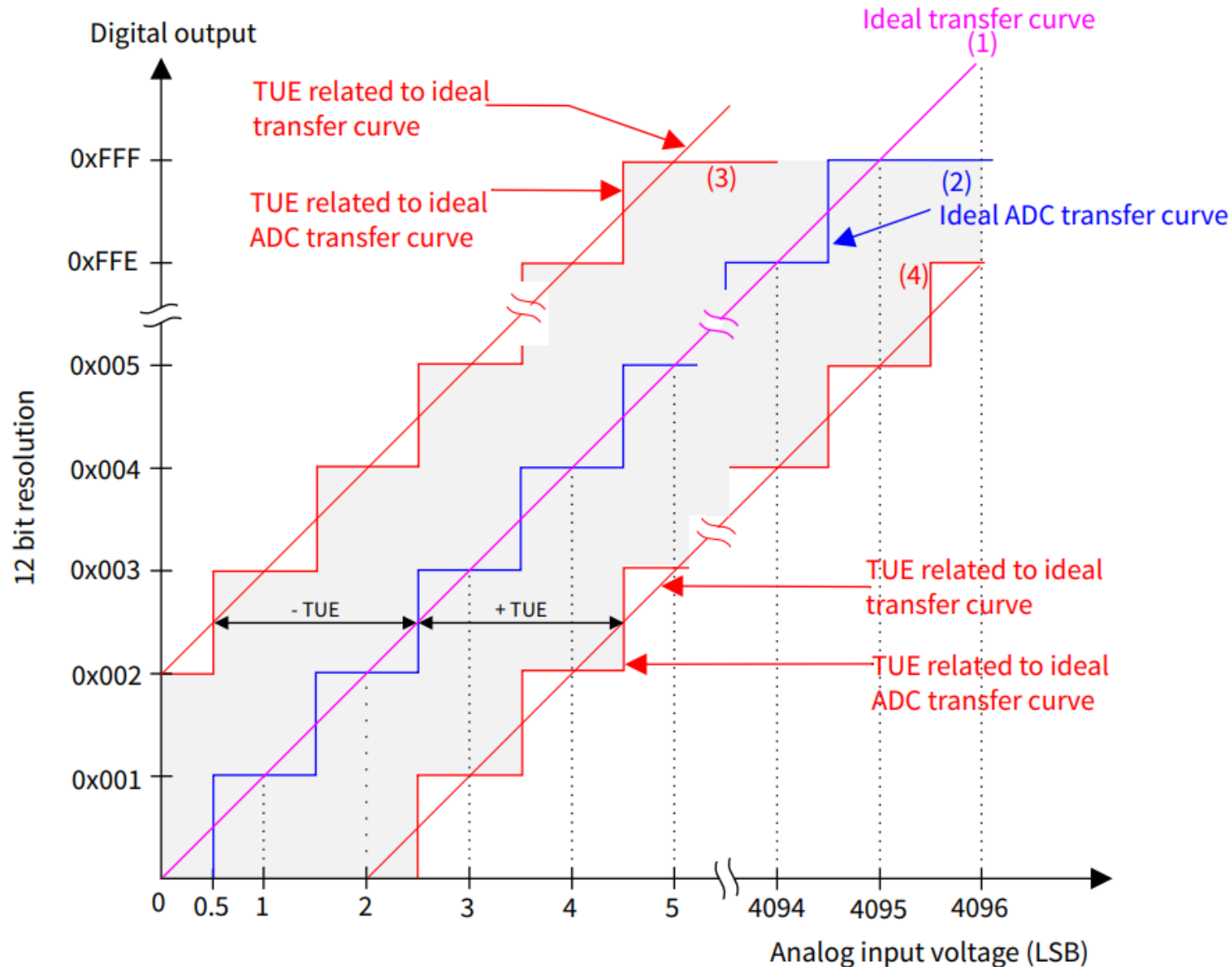
Result = 01111000

Decimal Result = 120

Expected Result = $(0.5 / 1) \times 255 \approx 127$

Error is approximately 7 LSBs for an 8-bit ADC → 256 Quantization Steps

SAR ADC – CONVERSION ERRORS



Voltage Resolution of a 12-bit ADC:

Max Range: 0xFFF = 4096

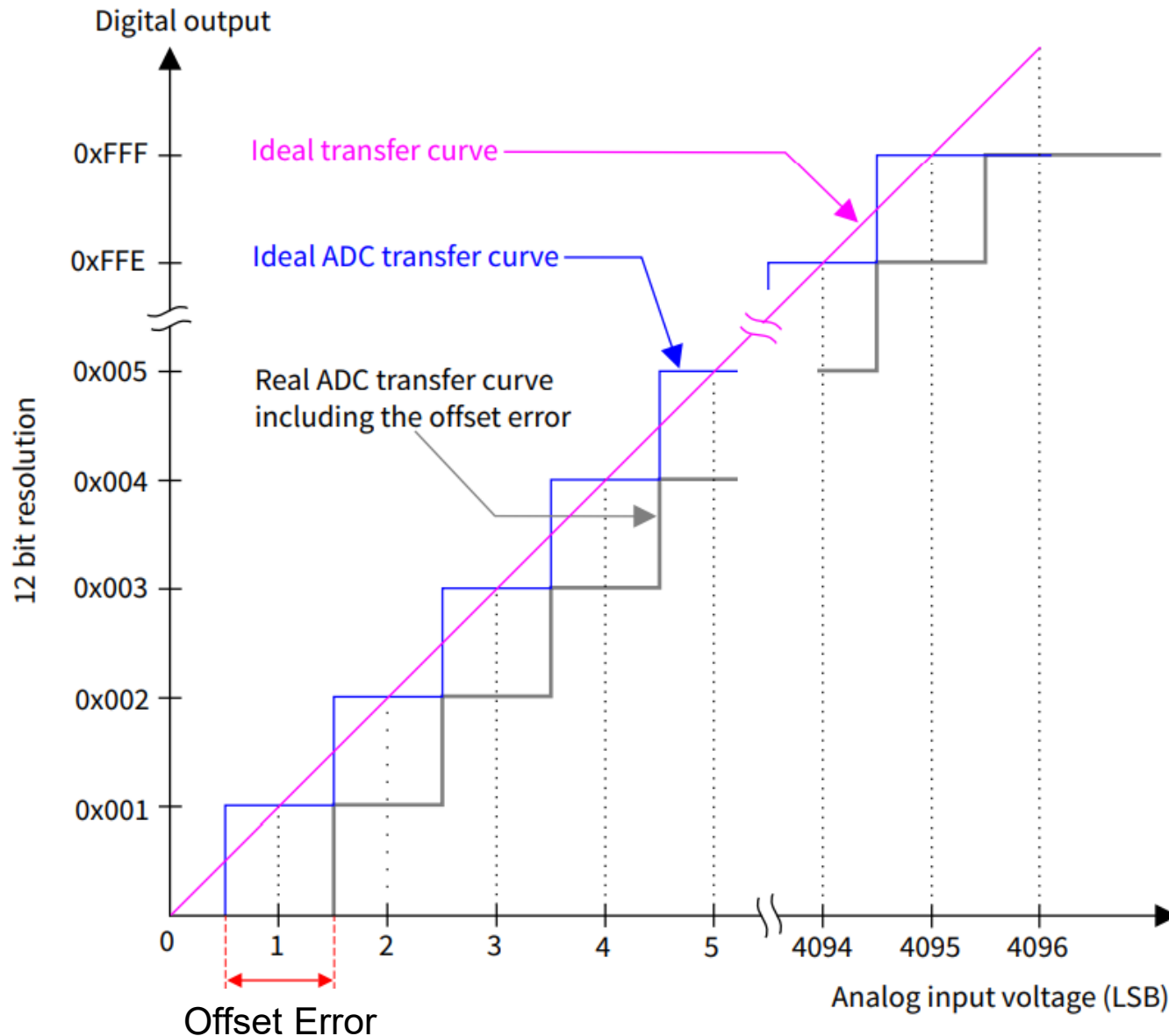
The analog input voltage range must be within VSS (GND), up to VREF. If VREF = 5.0 V, then

$$\text{Resolution} = 5.0 \text{ V} / (4095 \text{ LSBs}) = 1.22 \text{ mV/LSB}$$

TUE: Total Unadjusted Error → Total Impact of all possible errors that can arise during an AD conversion without any prior calibration:

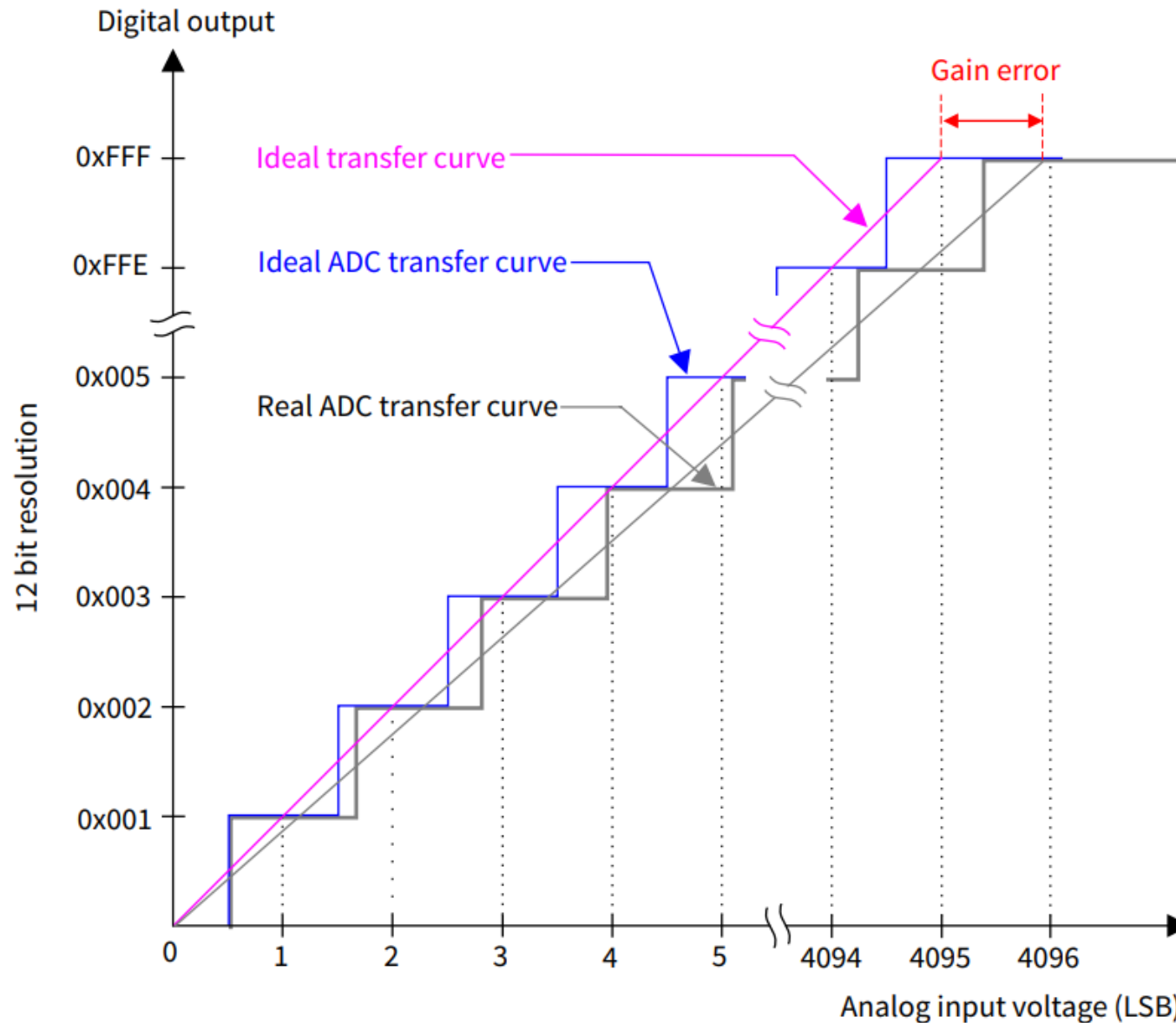
- ❖ Quantization Error
- ❖ Offset Error (Op amps, component tolerances etc.)
- ❖ Gain Error
- ❖ Differential Nonlinearity (DNL) Error
- ❖ Integral Nonlinearity (INL) Error

SAR ADC – CONVERSION ERRORS



The offset error is the deviation from the Ideal ADC Transfer Curve at the lowest transition level on the Real ADC Transfer Curve. It is the input voltage required to bring the digital output to zero and can be measured by determining the first digital transition, from 0 to 1, of the ADC. The offset error affects all codes by the same amount.

SAR ADC – CONVERSION ERRORS



The gain error is the difference between the slopes of the Real ADC Transfer Curve and the Ideal ADC Transfer Curve at the maximum digital out value.

SAR ADC – DIFFERENTIAL NONLINEARITY (DNL)

$$\text{Number of Codes} = 2^N$$

$$V_{LSB} = \frac{FSR}{2^N}$$

Where

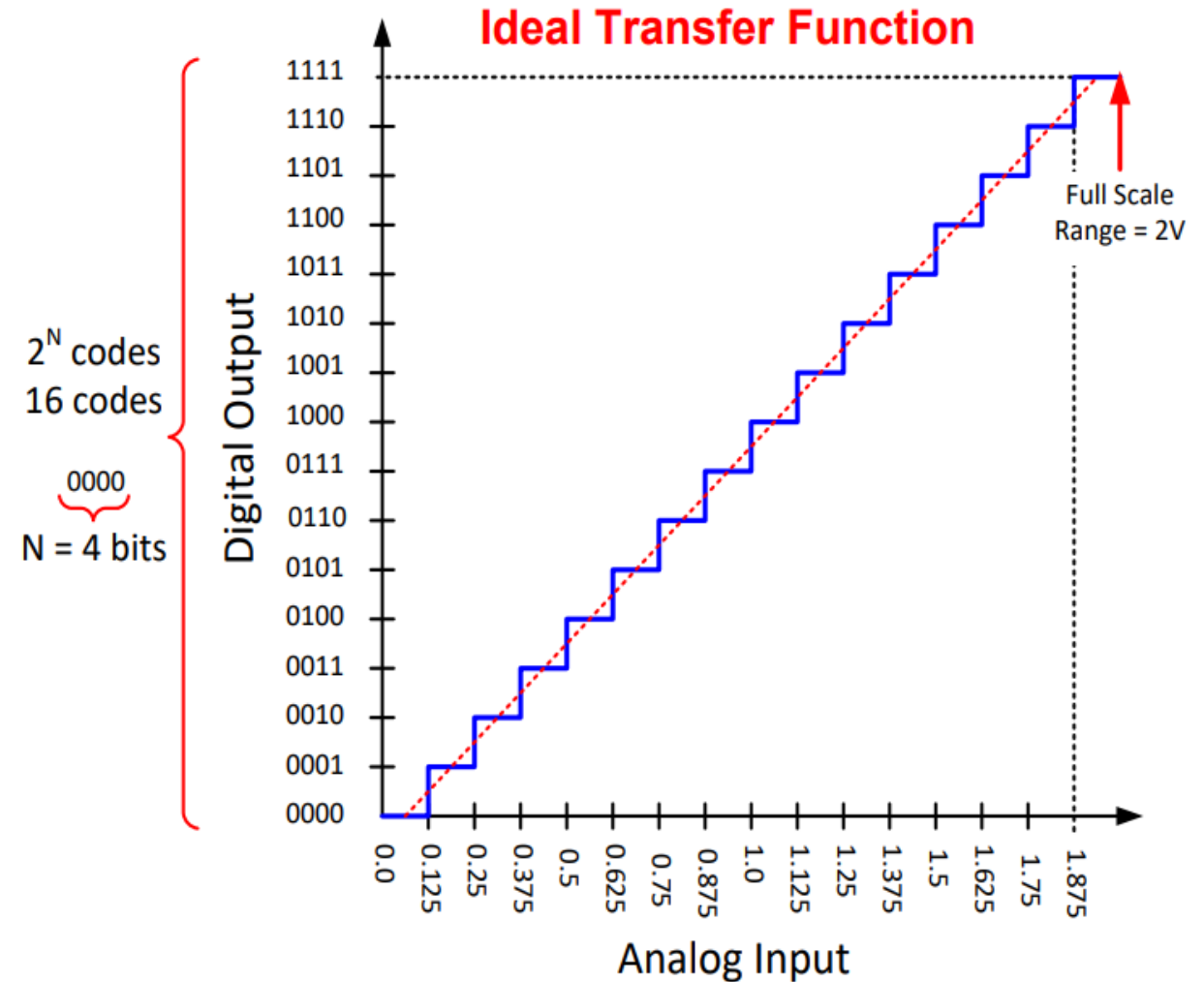
V_{LSB} = The minimum resolvable voltage width

FSR = Full Scale Range

N = Number of bits

$$V_{LSB} = \frac{FSR}{2^N} = \frac{2V}{2^4} = 0.125V$$

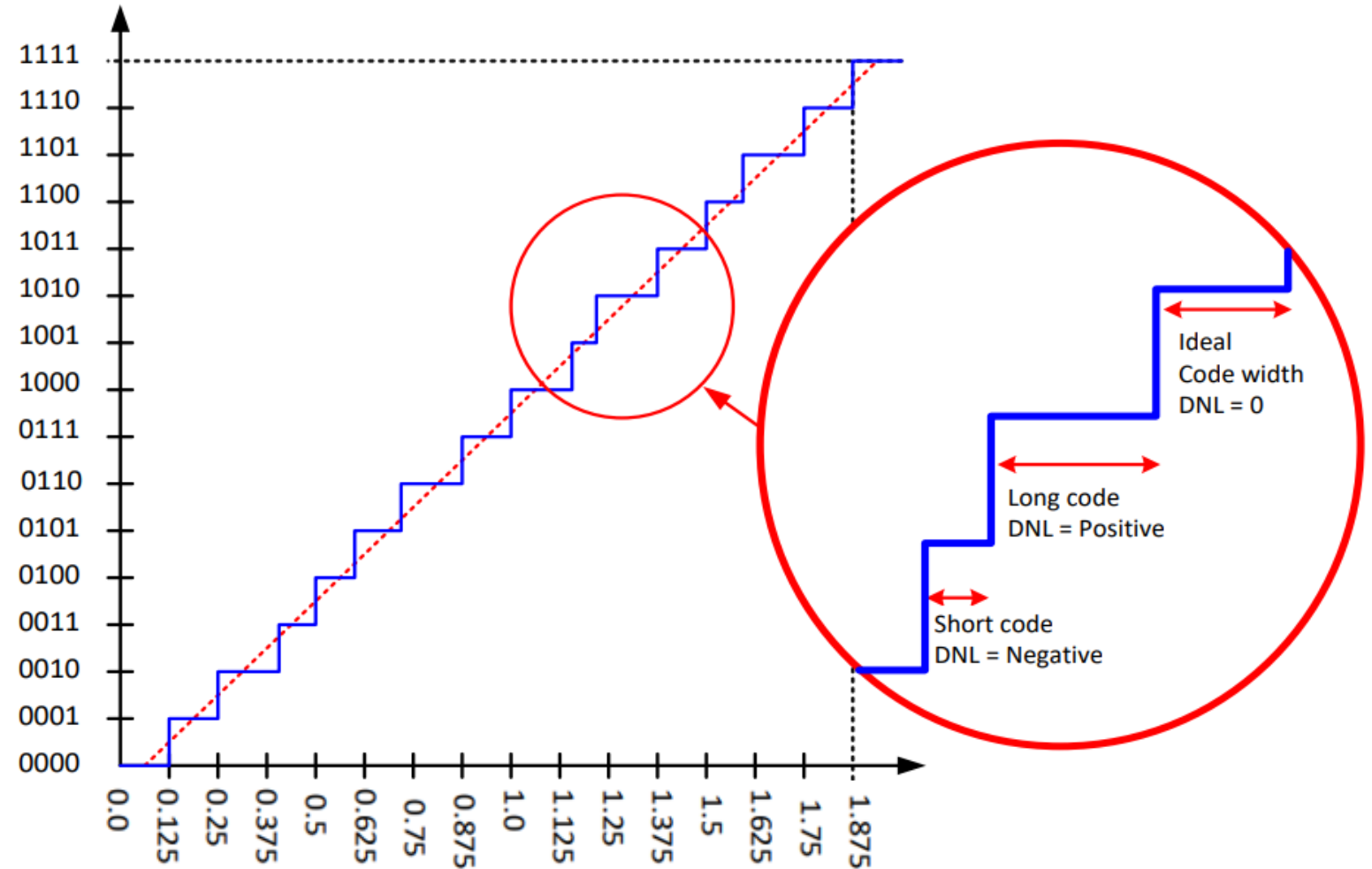
$$\text{Number of Codes} = 2^N = 2^4 = 16$$



SAR ADC – DIFFERENTIAL NONLINEARITY (DNL)

$$DNL[k] = \frac{W[k] - Q}{Q}$$

- $T[k]$: Input voltage at which the ADC transitions from code k to $k+1$
- $W[k] = T[k+1] - T[k]$: Measured width of code k
- Q : Ideal code width (1 LSB)



SAR ADC – DIFFERENTIAL NONLINEARITY (DNL)

Example 1:

ADC resolution: 4 bits

Number of codes: $2^4=16$

Full-scale input range: 0 – 1.6 V

Ideal Code Width (LSB)

$$Q = \frac{1.6 \text{ V}}{16} = 0.1 \text{ V}$$

For an ideal ADC:

Code k	$T[k]$ (V)	$T[k + 1]$ (V)	$W[k]$ (V)
5	0.50	0.60	0.10

$$\text{DNL}[5] = \frac{0.10 - 0.10}{0.10} = 0$$

Ideal Code Width – DNL = 0

Example 2:

$$T[6] = 0.60 \text{ V}$$

$$T[7] = 0.73 \text{ V}$$

$$W[6] = 0.73 - 0.60 = 0.13 \text{ V}$$

$$\text{DNL}[6] = \frac{0.13 - 0.10}{0.10} = +0.3 \text{ LSB}$$

The code is 30% wider than ideal

Example 3:

$$\bullet T[9] = 0.90 \text{ V}$$

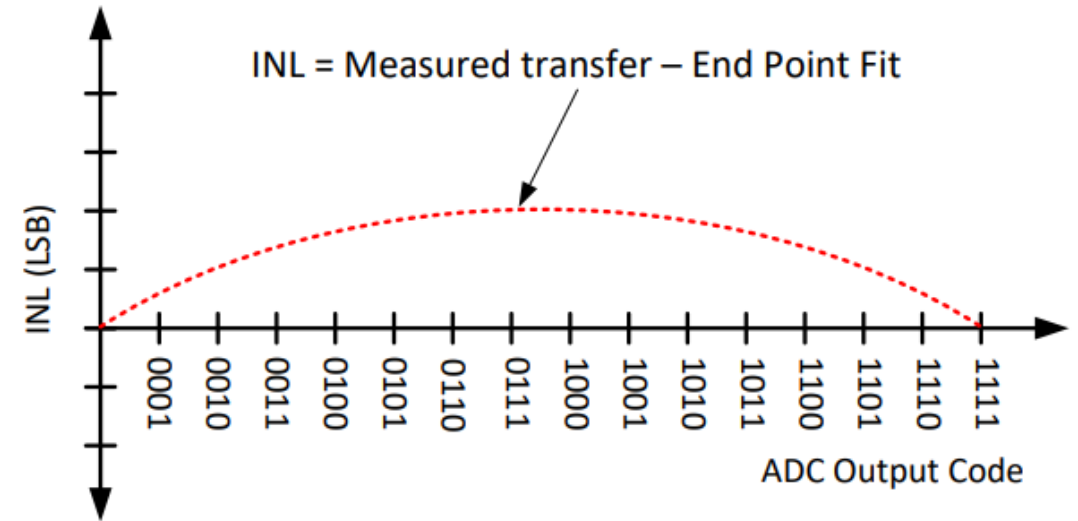
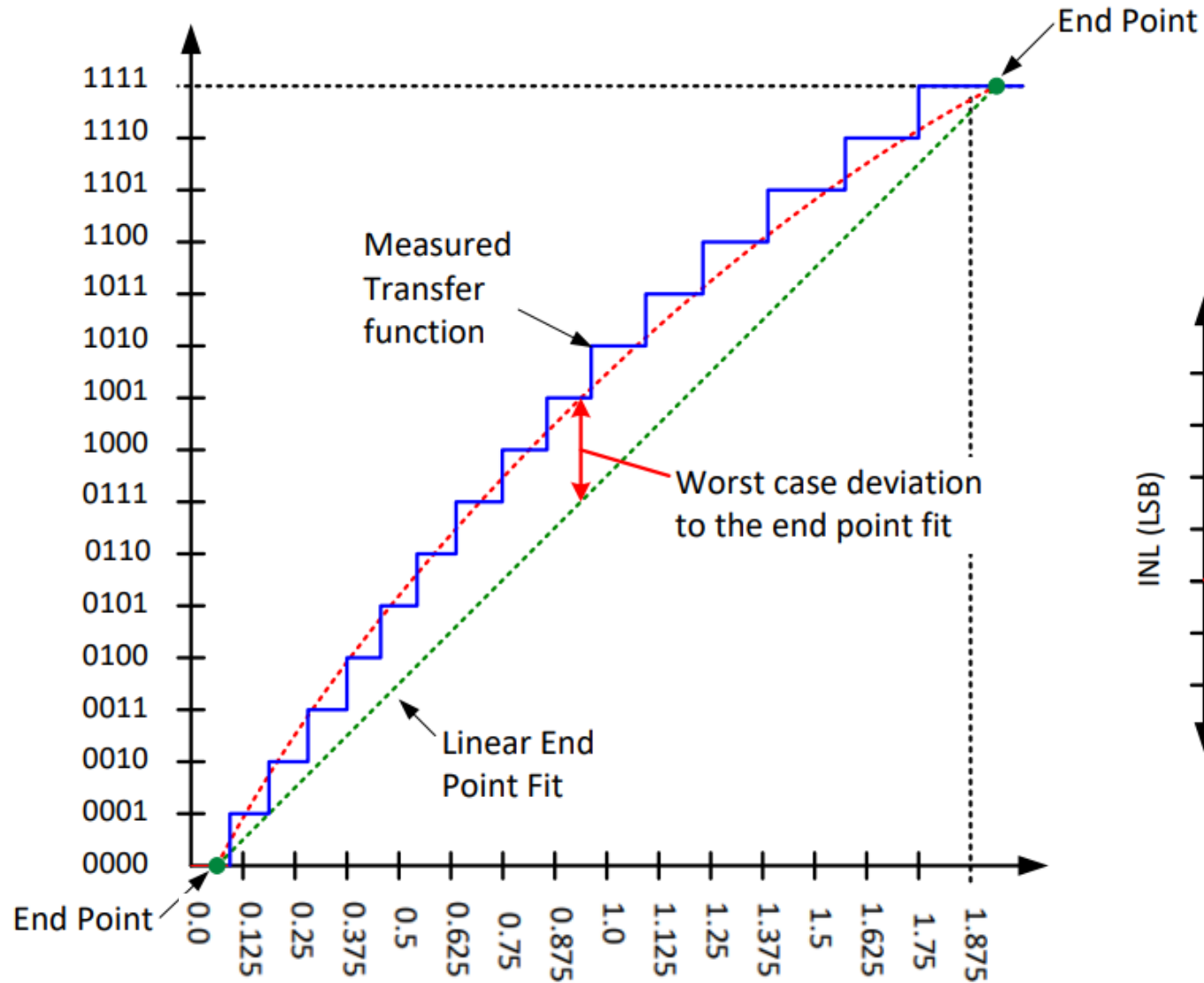
$$\bullet T[10] = 0.96 \text{ V}$$

$$W[9] = 0.06 \text{ V}$$

$$\text{DNL}[9] = \frac{0.06 - 0.10}{0.10} = -0.4 \text{ LSB}$$

The code is significantly narrower than ideal

SAR ADC – INTEGRAL NONLINEARITY (INL)



SAR ADC – INTEGRAL NONLINEARITY (INL)

Example:

$$\text{INL} = \text{Measured transfer} - \text{End-point fit}$$

- ADC resolution: 4 bits $\rightarrow$ 16 codes
- Full-scale voltage: 0 – 1.875 V (as shown on the x-axis)
- Ideal LSB:

$$Q = \frac{1.875}{16} = 0.1172 \text{ V}$$

- $T[0] = 0 \text{ V}$
- $T[15] = 1.875 \text{ V}$

The ideal transition voltage for code k under end-point fit is:

$$T_{\text{ideal}}(k) = k \cdot Q$$

$$T_{\text{measured}}(7) = 0.95 \text{ V}$$

$$T_{\text{ideal}}(7) = 7 \cdot 0.1172 = 0.820 \text{ V}$$

INL Calculation:

$$\text{INL}[7] = T_{\text{measured}}(7) - T_{\text{ideal}}(7)$$

$$\text{INL}[7] = 0.95 - 0.820 = 0.130 \text{ V}$$

$$\text{INL}[7] = \frac{0.130}{0.1172} \approx +1.11 \text{ LSB}$$

In this example, for $k = 7$, maximum error occurs. The maximum error can occur at a different code!

SAR ADC – TOTAL UNADJUSTED ERROR (TUE)

- ❖ DNL is caused by mismatch in the elements that set ADC code boundaries (resistors, capacitors, or current sources), so adjacent code steps are not exactly equal in width.
- ❖ INL is caused by systematic nonlinearities that accumulate across many codes, such as DAC nonlinearity, reference curvature, or stage gain errors, which bend the overall transfer curve.
- ❖ Offset error is caused by input-referred offsets in comparators, amplifiers, or the sampling front-end, shifting the entire transfer curve left or right.
- ❖ Gain error is caused by inaccuracies in the reference voltage or internal gain factors, changing the slope of the transfer curve relative to the ideal.

$$TUE = \sqrt{(OE^2 + GE^2 + DNL^2 + INL^2 + QE^2)}$$

TUE: Total Unadjusted Error

OE: Offset Error

GE: Gain Error

DNL: Differential Nonlinearity

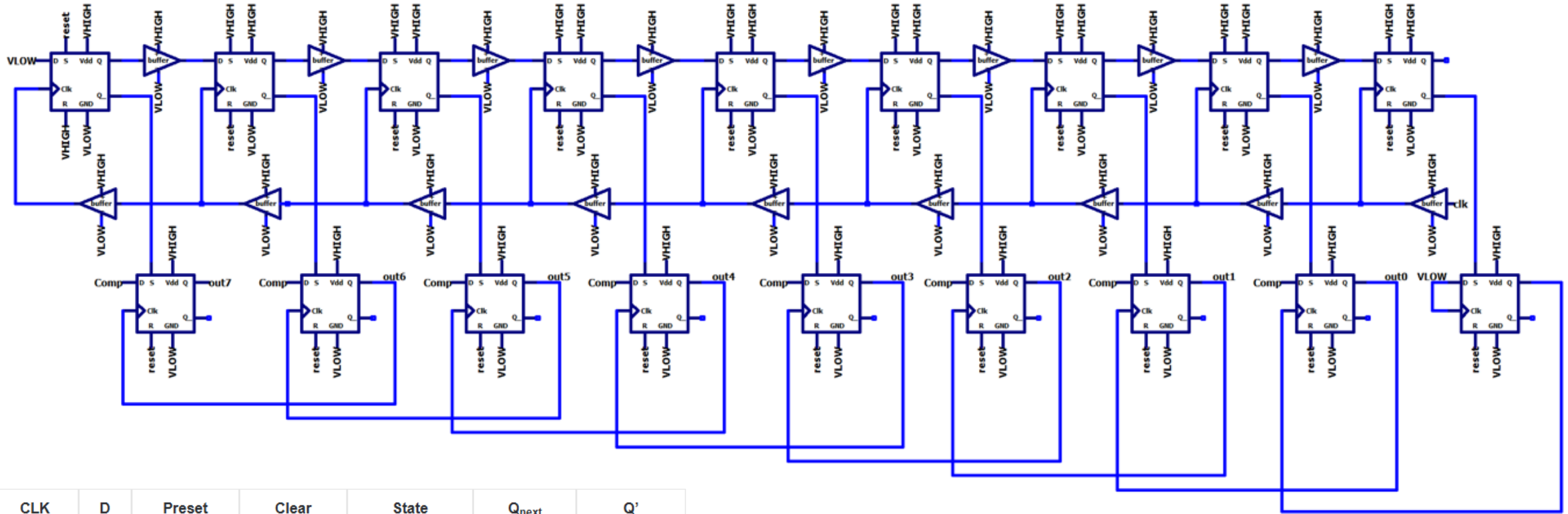
INL: Integral Nonlinearity

QE: Quantization Error

SAR ADC

ADDENDUM

SAR ADC – LOGIC & REGISTER



CLK	D	Preset	Clear	State	Q_{next}	Q'
X	X	1	0	Preset	1	0
X	X	0	1	Clear	0	1
X	X	0	0	Hold	Previous State	Previous State
Rising edge	0	0	0	Reset	0	1
Rising edge	1	0	0	Set	1	0

SAR ADC – LOGIC & REGISTER

The conventional SAR logic consists of two rows of shift registers and operates synchronously, assigning an equal conversion time to each conversion cycle. Its operation can be described as follows. At the start of conversion, the output of the first flip-flop in the upper shift register is set to logic '1', while its complementary output triggers the first flip-flop in the lower shift register. The upper shift register functions as a state machine, whereas the lower shift register serves as a memory to store the conversion results.

Once the first flip-flop in the lower shift register is activated, the initial comparison is performed and the comparator generates a decision. In the subsequent clock cycle, the second flip-flop in the upper shift register is activated, causing the output of the second flip-flop in the lower shift register to become logic '1'. Simultaneously, the clock input of the first flip-flop in the lower register is asserted to latch the result of the first comparison. This sequence continues until all bits have been resolved. After the final conversion step, both registers are fully reset in preparation for the next conversion cycle.

This architecture represents the fundamental SAR logic structure and is well suited for low- to medium-speed ADCs, as the propagation delay of the SAR logic ultimately limits the maximum achievable conversion speed.

References

1) www.halbleiter.org

The Principle of Semiconductor, Tokyo Electron Ltd.

2) Paul Gray, Robert Meyer, "Analysis and Design of Analog Integrated Circuits", John Wiley and Sons

3) Phillip E. Allen, Douglas R. Holberg, "CMOS Analog Circuit Design", Oxford University Press, 2002.

4) Texas Instruments Application Note, "Choose the right A/D converter for your application".